

0 7

Scientists investigated changes in the mass of fish from three populations of the same species. The fish they used had a life cycle of one year.

The scientists set up three fish tanks, each containing a separate population. Each year the scientists removed all the fish from each tank and determined the mean mass of the fish removed. They then put back 10% of each population in the following way.

Tank **A** – put back only the largest fish.

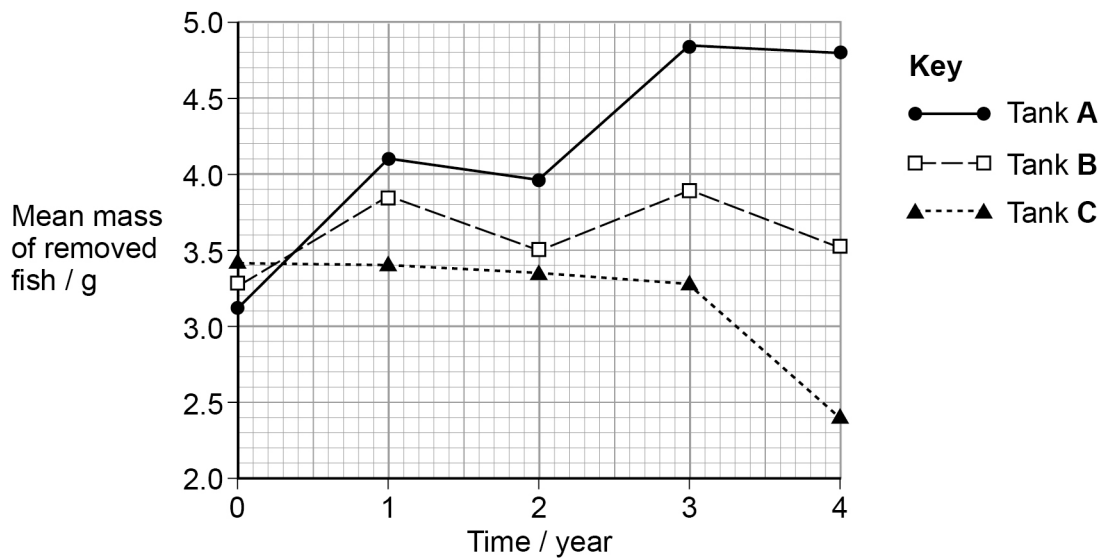
Tank **B** – put back fish at random.

Tank **C** – put back only the smallest fish.

During each year the fish were left to grow and reproduce.

The scientists' results are shown in **Figure 8**.

Figure 8

**0 7 . 1**

What type of selection were the scientists modelling in this investigation by putting back only the largest or only the smallest fish in Tank **A** and Tank **C**? Give a reason why.

[2 marks]

Type of selection _____

Reason _____

Question 7 continues on the next page

Turn over ►



0 7 . 2 Explain the purpose of Tank **B** (on page 21).

[2 marks]

0 7 . 3 Calculate the ratio of the mean mass of fish removed from Tank **A** (on page 21) to the mean mass of fish removed from Tank **C** (on page 21) at 1 year **and** at 4 years.

How much greater is the ratio at 4 years compared with the ratio at 1 year?

[2 marks]

Ratio at 1 year = _____

Ratio at 4 years = _____

How much greater at 4 years = _____



Sea fishing is regulated in law. The size of the mesh used in some fishing nets is controlled so that small fish can escape but large fish are captured. This regulation is designed to protect populations of wild fish.

Using all the information in this question, evaluate whether the scientists' investigation supports the use of these types of nets in sea fishing.

[3 marks]

[illegible]

[Extra space]

9

Turn over for the next question

Turn over ►

